

# 02YMECH - Solution 6

Assigned for the week of Oct 27, 2025

## Solutions

### 1. Phase-space diagrams

- **Left diagram:** Trajectories form closed orbits around the origin. Motion is bounded in time. The equilibrium at the origin is *stable*.
- **Right diagram:** Trajectories move away from the origin. The equilibrium at the origin is *unstable*.

### 2. Particle motion in polar coordinates

Given

$$r(t) = r_0 + v_r t, \quad \theta(t) = \Omega t + \frac{1}{2} \alpha t^2.$$

(a) General expressions:

$$\begin{aligned} \vec{v} &= \dot{r} \hat{r} + r \dot{\theta} \hat{\theta}, \\ \vec{a} &= (\ddot{r} - r \dot{\theta}^2) \hat{r} + (r \ddot{\theta} + 2 \dot{r} \dot{\theta}) \hat{\theta}. \end{aligned}$$

(b) Components:

$$\begin{aligned} v_r &= v_r, & v_\theta &= (r_0 + v_r t)(\Omega + \alpha t), \\ a_r &= -(r_0 + v_r t)(\Omega + \alpha t)^2, \\ a_\theta &= (r_0 + v_r t)\alpha + 2v_r(\Omega + \alpha t). \end{aligned}$$

(c) Speed and its derivative:

$$\begin{aligned} v(t) &= \sqrt{v_r^2 + (r_0 + v_r t)^2 (\Omega + \alpha t)^2}, \\ \dot{v} &= \frac{(r_0 + v_r t)(\Omega + \alpha t) [\alpha(r_0 + v_r t) + v_r(\Omega + \alpha t)]}{v(t)}. \end{aligned}$$

The sign of  $\dot{v}$  is determined by the numerator.

(d) Radial acceleration vanishes when

$$a_r = 0 \Rightarrow t^* = -\frac{r_0}{v_r} \quad \text{or} \quad t^* = -\frac{\Omega}{\alpha}.$$

Such times exist for  $t^* \geq 0$  if  $v_r < 0$  and/or  $\Omega\alpha < 0$ .